Corn Yield Prediction

Crop Yield Forecasting using Prithvi AI and Ridge Regression

CSU Hackathon 2026 | USDA NASS Challenge (Prompt 2)

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Team 13: “Doomed from the Start”

THE PROBLEM

OUR MISSION: Build a geospatial AI pipeline to forecast corn-for-grain yields at 4 key points during the 2025 growing season across 5 states.

"If USDA yield estimates are wrong, it's like giving the entire food economy bad GPS directions. Everyone follows them and the system ends up in a bad place."

Price Stability

Drives commodity market decisions globally

Supply Chain

Enables planning across the food economy

Farm Decisions

Informs farm-level strategy & confidence

National Impact

USDA NASS estimates relied upon nationally

TARGET STATES & FORECAST TIMELINE

COVERED STATES

- Iowa
- Colorado
- Wisconsin
- Missouri
- Nebraska

FOUR FORECAST WINDOWS

01
AUG 1
Early Grain Fill

02
SEP 1
Late Grain Fill / Dough Stage

03
OCT 1
Maturity / Early Harvest

04
END OF SEASON
Post-Harvest Reconciliation

OUR PROCESS

STEP	TOOL / METHOD	OUTCOME
1. Define Scope & Acquire Data	USDA NASS, NASA HLS	Historical yield data + satellite imagery pulled for Iowa corn counties
2. Identify Corn Fields	CDL Masking, NDVI Analysis	Filtered satellite pixels to corn-only fields using Cropland Data Layer
3. Preprocess & Engineer Features	SageMaker (goobie6000.ipynb)	Spectral bands, NDVI time series, and weather features prepared for model input
4. Train Prithvi-EO Model	NASA Prithvi-EO, AdamW, GPU	Fine-tuned ViT on NASS 2005–2024 Iowa yield data across 120 epochs
5. Use Prithvi Output in Regression Model	XGBoost, RandomForestRegressor, Data Regularization	Regression model for Corn Predictions by Month
6. Generate Forecasts & Uncertainty Cones Using Regression Model	Analog Year Matching, Quick Suite	Iowa yield forecasts at 4 seasonal windows with confidence bounds

All preprocessing and model training conducted in goobie6000.ipynb on AWS SageMaker · Iowa corn counties · 2025 growing season

CHALLENGES

Cloud Cover & Data Gaps

Data Quality

HLS imagery is frequently obscured by clouds during critical growth stages, requiring gap-filling strategies and temporal interpolation across Landsat and Sentinel-2 passes.

Cross-State Variability

Generalization

Soil types, irrigation practices, and local climates differ significantly across IA, CO, WI, MO, and NE — requiring the model to generalize across highly varied agronomic conditions.

Analog Year Selection

Methodology

Identifying truly comparable historical years is non-trivial. Climate patterns interact with crop genetics and market factors, making simple weather-matching insufficient.

Ground Truth Scarcity

Scale Mismatch

USDA NASS reports county-level yields annually — not field-level, not in-season. Bridging coarse ground truth to high-resolution satellite pixels required careful spatial aggregation.

GPU Unavailability & CPU Fallback Cost

Infrastructure

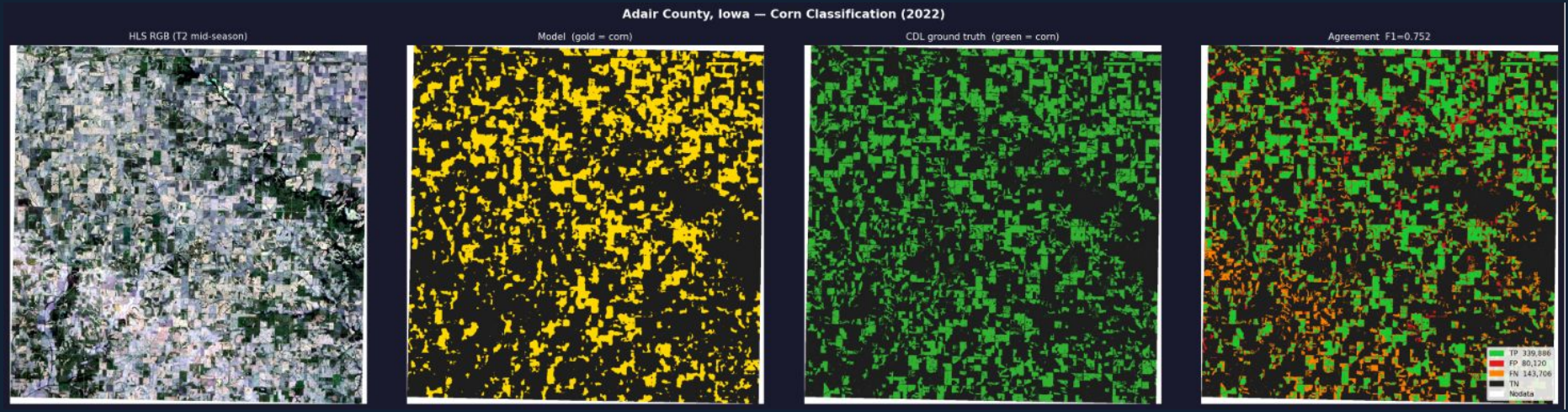
GPU access on SageMaker was blocked during training, forcing us to fall back to CPU—costing several hours of lost time. Running a Vision Transformer on CPU is significantly slower, which limited the number of training iterations we could complete within the hackathon window.

In-Season Label Availability

Validation

Final NASS yield estimates aren't published until after harvest, making real-time validation impossible mid-season. We relied on held-out historical years as a proxy.

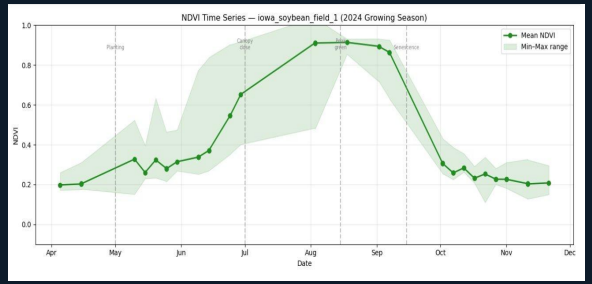
CORN CLASSIFICATION



INITIAL STEPS: IDENTIFYING CORN

Our first task was understanding what satellite data looks like for corn vs. other crops. We started with NDVI time series and raw spectral band visualizations to establish a baseline phenological signature for corn.

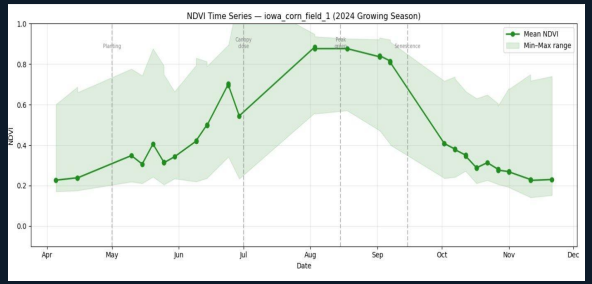
Step 1: Soybean NDVI Baseline



Soybean as Reference

Plotting soybean NDVI first gave us a reference phenological curve. The shape, timing, and peak helped us distinguish it from corn signatures in later analysis.

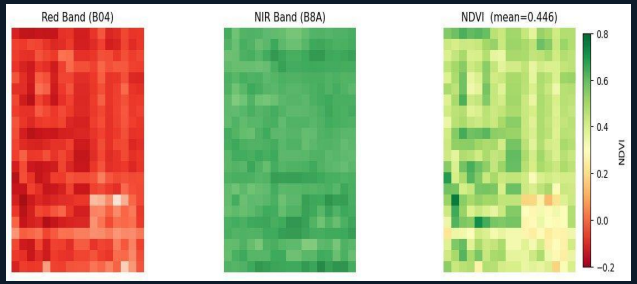
Step 2: Corn NDVI Time Series



Corn NDVI Phenology

Corn NDVI peaks in late summer to fall (Aug–Sep) but closely mirrors soybean patterns, so NDVI alone was not sufficient to distinguish between crops.

Step 3: Spectral Bands (B04 / B8A / NDVI)



Red + NIR → NDVI

Visualizing raw B04 (Red) and B8A (NIR) Sentinel-2 bands confirmed field-level variation. NDVI mean=0.446 indicated moderate early-season greenness consistent with pre-canopy closure.

OUR DATA SOURCES

DATASET	SOURCE	ROLE
HLS Multispectral Imagery (Landsat + Sentinel-2)	NASA	Vegetation indices & crop health signals
NAIP Aerial Imagery	USDA FSA (via AWS)	High-res field boundaries & land cover
Cropland Data Layer (CDL)	USDA NASS	Corn field identification & masking
Historical Corn Yield Data (1941–2024)	USDA NASS QuickStats	Ground truth for training & validation
Weather / Climate Data	NOAA / NASA	Analog year identification & feature engineering

Data ingestion and preprocessing handled in goobie6000.ipynb on AWS SageMaker

MODEL ARCHITECTURE

FOUNDATION MODEL

- NASA open-source geospatial AI foundation model
- Pre-trained on large-scale HLS satellite imagery
- Vision Transformer (ViT) architecture
- Optimizer: AdamW · Training: 86 epochs
- Fine-tuned on NASS corn yield data (1941–2024)

PIPELINE FLOW

Raw Satellite Imagery

HLS Landsat/Sentinel-2 + NAIP

▼

Preprocessing & CDL Masking

Corn Fields Only

▼

Prithvi-EO Feature Extraction

Vision Transformer

▼

Yield Regression Head

Fine-tuned on NASS 1941–2024

▼

State-Level Yield Forecast

With Uncertainty Cone

2025 SEASON RESULTS

Iowa	Colorado	Wisconsin	Missouri	Nebraska
Forecasts Generated	Forecasts Generated	Forecasts Generated	Forecasts Generated	Forecasts Generated
✓	✗	✗	✗	✗

KEY HIGHLIGHTS

1
State Covered
IOWA

1
Forecast Points
End of Season

83
Training Years
USDA NASS 1941–2024

86
Training Epochs
AdamW Optimizer

Dashboards and visualizations built and connected via Quick Suite.

MODEL RESULTS

Output from Fine-Tuned Prithvi Model

```
============================================================
  CORN CLASSIFICATION RESULTS — POLK COUNTY, IOWA (FINE-TUNED)
============================================================

  Pixel-level metrics (corn vs. non-corn):
    True Positives  :    278,161  px  (corn correctly identified)
    False Positives :     45,485  px  (predicted corn, actually not corn)
    False Negatives :     63,114  px  (corn missed by model)
    True Negatives  :  1,314,662  px  (non-corn correctly identified)

  Performance:
    Precision (corn) : 0.859
    Recall    (corn) : 0.815
    F1 Score  (corn) : 0.837
    IoU       (corn) : 0.719
    Pixel accuracy   : 0.936

  Area estimates:
    Model predicted corn :     71,977 acres
    CDL ground-truth corn:     75,898 acres
    Area difference      : -   3,921 acres (-5.2%  undercount)
```

MODEL RESULTS

Output from Regression Model

```
Trend for 2022                  : 192.0 bu/acre
  Final yield prediction            : 194.0 bu/acre
  Actual NASS yield (2022)      : 188.6 bu/acre
  Absolute error                    : 5.4 bu/acre
  Relative error                    : 2.9%
  ✅ Excellent — within 5% of NASS ground truth
  Predicted corn acres              : 72,006 acres
  Estimated total production        : 13,971,702 bushels
```

INNOVATION & SCALABILITY

WHAT MAKES US INNOVATIVE

Prithvi-EO Foundation Model

Cutting-edge NASA model vs. traditional regression approaches

Multi-Resolution Fusion

High-res NAIP combined with multi-temporal HLS for richer spatial context

OPERA Integration

Surface water & land disturbance products for environmental risk signals

Data-Driven Uncertainty

Analog-year cones — operationally interpretable confidence intervals

SCALABILITY

Multi-Crop Ready

Modular pipeline extends to soybeans, wheat, and beyond

Cloud-Native Deployment

Easily deployable to additional states via SageMaker

Annual Reusability

Reusable year-over-year with updated NASS and satellite data

WHY THIS MATTERS

Impact for USDA & NASA

USDA NASS

Directly supports the mission of timely, accurate agricultural statistics

Applications

Useful for both individual farmers and large-scale agricultural statistics

NASA Earth Observation

Demonstrates how NASA Earth observation assets can be used to predict agronomic trends with great precision

Cutting Edge Machine Learning Techniques

Usage of cutting edge machine learning models and techniques including random forest classifiers and geospatial transformers

Thank You

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Questions?

BUILT WITH

- NASA Prithvi-EO
- HLS Landsat/Sentinel-2
- NAIP
- USDA NASS
- NOAA Weather
- SageMaker
- Quick Suite